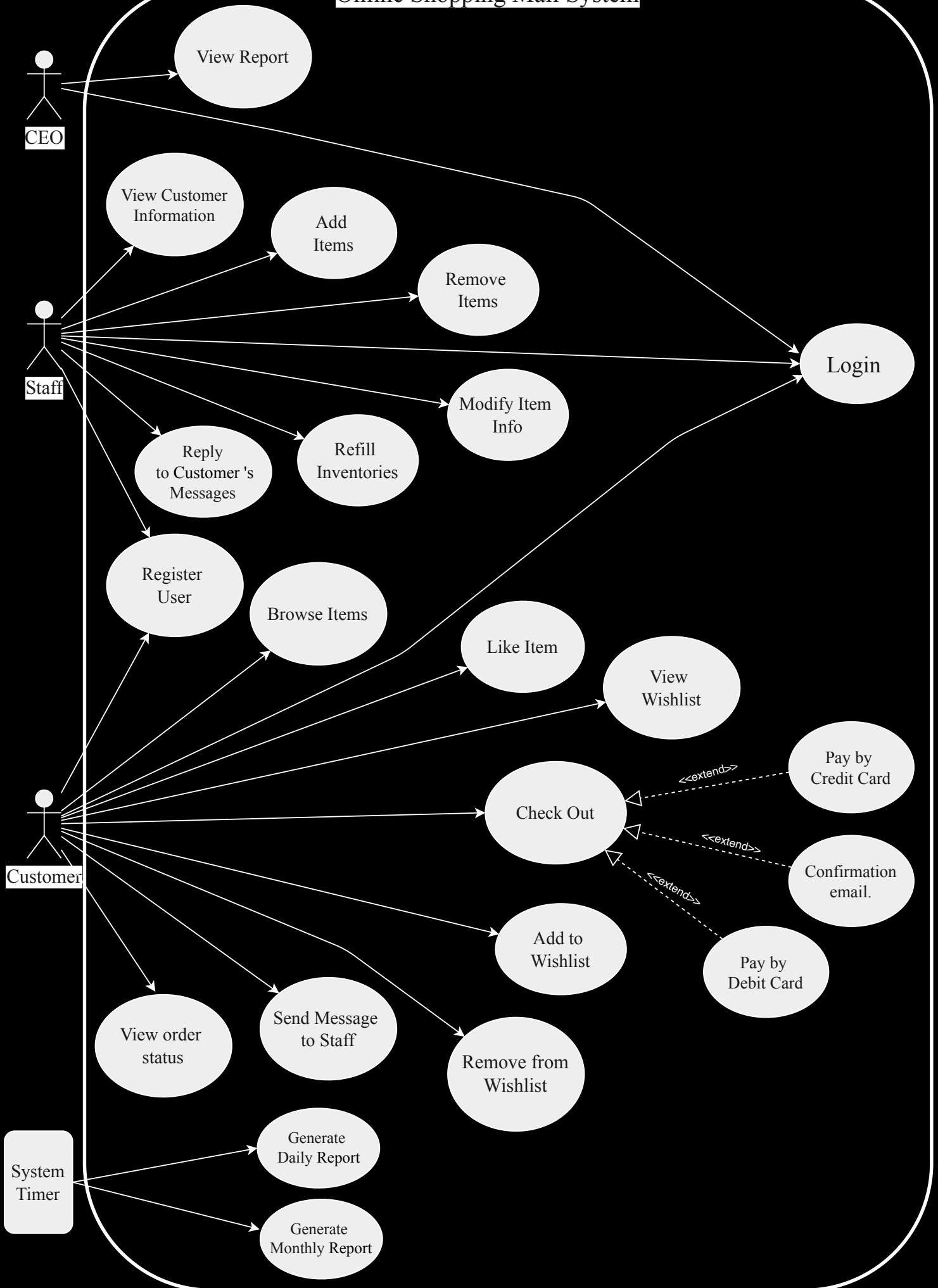


Online Shopping Mall System



Phase 1

Use Case Name: Login

Actors: Customer, Staff, CEO

Summary: Users are presented with a log in screen where they must enter a username and password to gain access to the shopping mall. Access rights are determined by the user's role. Customers access the shopping mall, Staff members manage inventory and access customer information, and the CEO views financial reports for daily, and monthly transactions.

Precondition: The user must be registered as customer, staff, or CEO.

Description of main sequence:

1. The user selects the login option on the welcome screen.
2. The user inputs their username and password into the corresponding fields on the login screen.
3. The system retrieves the entered credentials.
4. System verifies the entered credentials against registered user information
5. The system confirms the user account is valid and grants access to the shopping mall.

Postcondition: Credentials are accepted; users are granted access to online shopping mall, and the system navigates to the home screen.

Alternative sequences:

Step 3: Unregistered customer: If a customer does not currently have a valid account the system will redirect them to the “Register User” page to create an account and become a registered user.

Step 3: A Staff or CEO’s credentials are not found in the system: If a staff member or CEO forgets their credentials the System directs them to a new page for forgotten credentials.

Step 4: Credentials not matched: If the credentials do not match with the registered users, the system asks users to try again or register.

Step 4: Too many invalid passwords: If there are more than 5 wrong password attempts then the system locks the user and asks to try again after 10 mins.

Use Case Name: Register User

Summary: A new user submits personal information to create an account in the system.
Registration is required to access the shopping mall application.

Actors: Customer, Staff

Precondition: The system is online and displaying the welcome screen and the user is not registered in the system.

Description of main sequence:

1. The user selects the “Register” option on the welcome screen.
2. The system prompts the user to enter required details (e.g., name, email address, password).
3. The user enters the required data.
4. The system asks users if they are staff or customer, if staff is selected then the system asks an additional staff code to confirm that.
5. The System validates the entered data (e.g., checks email format, ensures password meets security requirements).
6. The system creates a new user account with the provided details.
7. The system redirects the user to the main application dashboard.

Postcondition: A new user account is created in the system, and the user is authenticated (logged in) and directed to the main application.

Alternative sequences:

Step 4 : Invalid staff code: If the entered staff code is wrong then, the system prompts a wrong code message and returns to step 2.

Step 5: Invalid Data Input: If the user inputs invalid data, the system displays an error message and returns to step 2, giving the user the opportunity to re-enter their information.

Step 5: The User Cancels during the validation process (for example, they exit the page):
If at any point during steps 2-4, the system discards the entered data, returns to the welcome screen, and ends the registration process.

Step 5 : Email already registered: If the email is already registered, the system prompts the user to enter a different email.

Use Case Name: Browse Items

Actors: Customer

Summary: This page allows the customer to view items available for purchase in the online shopping mall. Items are sorted by popularity based on the customers' liked items.

Precondition: Customers must be logged into the system and there must be items in the catalog.

Description of main sequence:

1. Customers are directed to the “Browse Items” page.
2. The system displays the available items in order according to the number of likes.
3. Customers are able to view the list of available items.
4. Customers can select items and view item details.

Postcondition: Customers can browse the list of available items in the online shopping mall.

Alternative sequences:

Step 2: No items available: If a customer navigates to the “Browse Items” page and there are no items available in stock then the system will display the message “No Items Currently Available”. The user is able to navigate out of the “Browse Items” page.

Step 4: Item details fail to load: If a customer selects an item but the system fails to load the item details it will display an error message “Unable to Load Item Details. Try Again Later”. The customer remains on the “Browse Items” page.

Step 4: Item no longer available: If an item becomes unavailable after it was displayed to the customer, the system will notify the customer “This Item is no longer available” and return to the updated browsing list.

Use Case Name: Like Items

Actors: Customer

Summary: Customers have the option to “Like” an item after selecting it which increases its popularity. Liked items with increased popularity are the first to be displayed on the browsing page.

Precondition: The customer has selected an item from the “Browse Items” page.

Description of main sequence:

1. The customer clicks to select an item from the browsing page.

2. The customer selects the “Like item” option.
3. The system increases the item’s popularity score.
4. The system updates the browsing page to feature the most popular items in the catalog first.

Postcondition: The Customer likes a selected item and the system has increased its popularity score.

Alternative sequences:

Step 2: Customer is unable to like an item: Due to a system error, a customer is unable to like an item. The system does not add the item to the users liked items and displays the un-altered “Browse Items” page. The user is able to try again.

Step 2: Liking at item twice: The customer attempts to like the same item twice. The system recognizes this action and marks it as unliked. And if the user click the like button the system likes the item.

Use Case Name: View Wishlist

Actors: Customer

Summary: The customer will be able to select the “View Wishlist” option on the screen. After clicking the “View Wishlist” option the customer is presented with a list of every item they have added to their wishlist. The customer can select items and see item details.

Precondition: The customer must already have items added to the wishlist.

Description of main sequence:

1. The customer navigates to the “View Wishlist” page.
2. The system retrieves the customers wishlist data.
3. The system displays all of the items the customer has added to their wishlist.
4. The customer can select and view item details.

Post condition: The customer is now able to view the items they have added to their wishlist.

Alternative sequences:

Step 3: The customer's wishlist is empty: If the customer navigates to the wishlist page but currently has no items in the wishlist the system displays a message “No items in Wishlist” and directs the customer back to the browsing page.

Step 3: Item no longer available: If a customer has added an item to their wishlist that has since been removed the system displays “This Item is no Longer Available”. The unavailable item is made inaccessible to the user so they can no longer select or view item details.

Use Case Name: Checkout

Actors: Customer

Summary: Customer reviews their order and selects payment information. The system confirms the order and sends an email receipt.

Precondition: The customer has selected items to purchase, added them to their wishlist, and initiated the checkout process.

Description of main sequence:

1. The user initiates the checkout process.
2. The system prompts the user to select a payment type.
3. The user selects a payment type and enters the payment information.
4. The system validates information and collects payment
5. The system generates an order record and receipt.
6. The system sends an order confirmation email with receipt to the customers registered email.
7. The user receives a confirmation email for their order with their order number.
8. The system displays the order confirmation message with its order number.

Postcondition: Customer’s order has been placed and a receipt has been sent to their registered email.

Alternative sequences:

Step 1: Inventory Changed During Checkout: An item in the customer's cart sells out while they are filling in payment details. The system identifies this, removes the item from the cart, informs the user with a message and recalculates the total before proceeding.

Step 3: Incomplete payment details: If a customer attempts to submit the payment information form without entering all of the required data the system detects the issue and prevents the form from submitting. The system highlights the required fields and displays a message “Required Fields are Missing”.

Step 4: Payment failed: The system fails to validate payment information and the order is not placed. The system displays an error message stating “Invalid payment

information please try again.” The system then allows the user to re-enter their payment information.

Step 6: Email Failed to send: If the system fails to generate and send the customer an order confirmation email the customer is notified with a message “Unable to Send Order Confirmation. View “Order Status” for more details or contact staff for assistance”.

Use Case Name: Confirmation email

Actors: Customer

Summary: After a successful transaction, the system sends an order confirmation email to the customers registered email address.

Dependency: Extends Checkout

Precondition: The customer must have completed an order with a successful payment.

Description of main sequence:

1. The system generates an order receipt after a successful payment.
2. The system then sends this receipt to the registered email address.
3. The system also displays the message of receipt sent.

Postcondition: The receipt has been sent to the customer’s registered email.

Alternative sequences:

Step 2 : Email not sent: Error occurred due to network connectivity issues and the system fails to send the order confirmation email to the customer. The system queues email for automatic retry and attempts to resend.

Use Case Name: Pay by Credit Card

Actors: Customer

Summary: Customer Inputs credit card information and the system validates the card and approves the method of payment.

Dependency: Extends Checkout

Precondition: Customer has initiated the checkout process and has chosen the credit card option when prompted to select payment type.

Description of main sequence:

1. The user initiates the checkout process
2. The system prompts the user to choose the payment type.
3. The user selects the 'Credit Card' option and enters credit card information.
4. The system validates information and collects payment.
5. The system displays the payment amount and confirmation.

Postcondition: The payment was successful, and the system sends the confirmation receipt to the customer's email.

Alternative sequences:

Step 4: Invalid card details: The credit card information is not valid. The system displays an inline message stating "Invalid Format" and highlights the field. The user remains on the payment screen and is allowed to re-enter the payment information.

Step 4: Payment could not be Collected: If the system fails to collect payment for an order (e.g. insufficient funds, issue with payment processor) it will display a message "Payment could not proceed. Try again later". The user remains on the payment screen and is able to try again, input another credit card, or cancel the transaction.

Use Case Name: Pay by Debit Card

Actors: Customer

Summary: Customer Inputs credit card information and the system validates the card and approves the method of payment.

Dependency: Extends Checkout

Precondition: Customer has initiated the checkout process and has chosen the debit card option when prompted to select payment type.

Description of main sequence:

1. The user initiates the checkout process
2. The system prompts the user to choose the payment type.
3. The user selects the 'Debit Card' option and enters credit card information.
4. The System validates information and collects payment
5. The system displays the payment amount and confirmation.

Postcondition: The payment was successful, and the system sends the confirmation receipt to the customer's email.

Alternative sequences:

Step 4: Invalid card details: The debit card information is not valid. The system displays an inline message stating “Invalid Format” and highlights the field. The user remains on the payment screen and is allowed to re-enter the payment information.

Step 4: Payment could not be Collected: If the system fails to collect payment for an order (e.g. insufficient funds, issue with payment processor) it will display a message “Payment could not proceed. Try again later”. The user remains on the payment screen and is able to try again, input another debit card, or cancel the transaction.

Use Case Name: Add item to Wishlist

Summary: Customers can browse the catalog and add desired items to their personal wishlist.

Actors: Customer

Precondition: The customer is logged into the system. The desired item is in stock and visible in the catalog.

Description of main sequence:

1. The customer browses the product catalog and selects a desired item.
2. The system displays the item details.
3. The customer selects the “Add to Wishlist” option.
4. The system checks the current stock level of the item.
5. If the item is in stock, the system adds the item to the customer's personal wishlist.
6. The system confirms to the customer that the item has been added to their wishlist.

Postcondition: The item is added to the customer's wishlist.

Alternative sequences:

Step 3: Item not available: The Item selected by the customer is no longer available because it was removed from the catalog. The system displays an “Item No Longer Available” message and does not add the item to the wishlist. The customer is allowed to continue viewing the catalog.

Step 5: Item Out of Stock: The "Add to Wishlist" option is disabled and the message "This item is currently out of stock and cannot be added to your wishlist" is displayed. The customer is allowed to continue viewing the catalog.

Use Case Name: Remove item from wishlist

Actors: Customer

Summary: The customer is able to manage their wishlist by removing items they no longer wish to purchase.

Precondition: The customer is logged into the system and has at least one item to their wishlist.

Description of main sequence:

1. The customer navigates to the “Wishlist” tab
2. The system displays all items currently in the customers wishlist.
3. The customer selects the “Remove item” option from the desired item.
4. The system prompts the user for confirmation.
5. The customer confirms the removal.
6. The system removes the item from the wishlist and displays a confirmation message stating the item was removed successfully.

Postcondition: The item is removed from the customer’s wishlist and the updated wishlist is displayed to the customer.

Alternative sequences:

Step 2: Wishlist is empty: The customer attempts to remove an item from an empty wishlist and the system displays “Wishlist Currently Empty” message and remains on the “Wishlist” page.

Step 4: Customer fails to confirm item removal: The customer does not confirm the removal of an item, the system does not remove the item, and returns to display the wishlist.

Step 5: System error: The system encounters an error when attempting to remove the item and displays a message asking the user to try again later. The item remains in the wishlist.

Use Case Name: Send Message to Staff

Summary: The customer can submit a written message to staff members describing their questions or issues.

Actors: Customer

Precondition: Customer has logged into the system.

Description of main sequence:

1. The customer selects the “Contact Staff” option.
2. The system provides a message composition form.
3. The customer writes and sends the message.
4. The system records the message and marks it as unread for staff review.

Postcondition: A new message from the customer is queued for staff review.

Alternative Sequences:

Step 3: Customer message fails to send: If the customer's message fails to send the system displays “Message not Sent. Try Again Later”. The customer remains on the message forum and the text is preserved so they can try again.

Step 3: Maximum words reached : The system limits the user to write the message at a certain words limit. If the user tries to exceed it, the system informs the user about the maximum limit reached.

Step 3: Minimum words not met : At Step 3, if the customer attempts to send a message that is below the minimum word limit , the system prevents the message from being sent and displays an alert message.

Step 3: Customer cancels the message: The customer does not want to send the message anymore and hits the cancel button. The system directs the customer back to the home page.

Step 4: The system does not record the customer’s message: If the system fails to record the customer's message it will preserve the customer's input and display “Your Message Could not be Saved”. The system queues the message for automatic retry.

Use Case Name: View Order Status

Actors: Customer, Staff

Summary: Allows the customer to view the status of their order. Customer order status is also available for staff review through their interface.

Precondition: The customer is logged in and must have placed an order for an item in order to view their order status.

Description of main sequence:

1. An order has already been placed by the customer.
2. The customer navigates to the view order status page.
3. The system prompts the customer to enter the order number.
4. The customer enters the order number.

5. The system retrieves and displays the order status. (e.g. Processing, Shipped, and Delivered)

Postcondition: The system displays the order status for a specific purchase.

Alternative sequences:

Step 4: The customer enters an invalid order number: The system shows an error message if the order number entered does not match any of the numbers in the system. The system will then prompt the user to enter the order number a second time and the flow returns to step 4.

Step 4: The order number is formatted incorrectly: If the customer enters an order number in the wrong format the system will display an inline message “Invalid Format” and remain on the “View Order Status” page. The customer is allowed to re-enter the order number.

Use Case Name: View Customer Information

Actors: Staff

Summary: Authorized staff views a customer's information such as personal details, order history, current orders, and payment information to assist them.

Precondition: The user is authenticated as a staff member during the log-in process

Description of main sequence:

1. Staff logs in to the system
2. Staff navigates to the “Customer Information” page
3. System prompts for customers email or username.
4. Staff enters the details.
5. The system searches for customers in the database and displays matching customer profiles.
6. Customer selects the customer.
7. Staff can review customer information or log out.

Postcondition: Staff has viewed the customer’s information.

Alternative sequences:

Step 5: No profile found: No matching profile is found for a given customer's credentials. The system displays a message stating “No Customer Found” and the staff member is prompted to re-enter the customer search criteria.

Step 6: No information for that customer: System is unable to retrieve or has no information for the selected customer.

Use Case Name: Add items to inventory/catalog

Actors: Staff

Summary: The staff is able to add new items to the inventory which makes them available for purchase.

Precondition: The user is authenticated as staff during the log-in process

Description of main sequence:

1. Staff navigates to the 'Manage Inventory' page.
2. System displays a list of inventory items including details
3. Staff clicks 'add item' button at the bottom
4. System prompts staff to enter the item information
5. Staff enters item information
6. System checks for duplicate item then adds new item to catalog

Postcondition: New items get added to the system and become available to customers for purchase.

Alternative sequences:

Step 5: No item information found: A staff member attempts to add an item without entering all of the required data. The system displays "Required Fields Cannot Be Left Blank" and prompts the staff member to re-enter the item information.

Step 8: Duplicate item: The system recognizes that the item already exists in the catalog and displays a "Duplicate Item" error message. Staff is prompted to modify the item details before proceeding.

Use Case Name: Remove Items from inventory/catalog

Actors: Staff

Summary: Allows authorized staff to remove an item from the catalog making it unavailable for customer purchase.

Precondition: There is at least one item in the catalog

Description of main sequence:

1. Staff navigates to the 'Manage Inventory' page.
2. The system displays a list of inventory items including details.

3. The staff clicks the 'delete item' button.
4. The system asks the user to confirm the item deletion.
5. Staff confirms deletion.
6. The system removes item from the catalog.

Postcondition: The item is removed from the catalog and from customers' wish lists.

Alternative sequences:

Step 5: Deletion not Confirmed: If the staff member is unable to confirm deletion the system cancels the operation and returns to the inventory list without removing the item.

Step 6: System error: Due to a system error the item cannot be deleted or confirmed for removal. The system displays an error message and the staff member is returned to the inventory list.

Use Case Name: Modify Item information

Actors: Staff

Summary: Allows authorized staff to view and modify inventory item information such as name, description, price, or category

Precondition: The user is authenticated as staff during the log-in process. At least one item exists in the Inventory.

Description of main sequence:

1. Staff navigates to the 'Manage Inventory' page.
2. The system displays a list of inventory items including details.
3. Staff clicks the desired item.
4. The system displays current detail for the item.
5. Staff modifies the values for one or more fields.
6. The system validates new data and updates.

Postcondition: The item information has been modified.

Alternative sequences:

Step 5: Cancel modification: If a staff member cancels the item modification all changes are discarded. The system returns to step 2 and displays the list of inventory items.

Step 5: Incorrect format: If a staff member enters item information of an incorrect format the system displays an inline error message "Incorrect Format" and does not

modify item information. The system remains on the “Manage Inventory” page and allows the Staff member to re-enter the item details.

Step 6: System error: System fails to update the item information and displays an “Unable to modify Details” error message. The system remains on the “Manage Inventory” page and continues to display original item information.

Use Case Name: Refill Inventories

Actors: Staff

Summary: Allows authorized staff members to restock items in the catalog that are running low, making them available for customer purchase.

Precondition: The user is authenticated as staff during the log-in process

Description of main sequence:

1. The Staff navigates to the ‘Manage Inventory’ page.
2. The system displays a list of inventory items including details. Items under the inventory threshold are highlighted.
3. Staff identifies items to restock and inputs desired quantity
4. System confirms restock operation

Postcondition: The selected item quantities are updated in the system and the items are available for customer purchase.

Alternative sequences:

Step 2: The Manage Inventory page does not display: If a staff member attempts to navigate to the “Manage Inventory” page and the system fails to load it, the system displays the “Could not Load Inventory. Try Again Later” message

Step 3: Low Stock is not highlighted: The system loads the inventory list as normal but fails to highlight low stock items. Staff can still restock inventory but must manually identify low-stock items or reload the page.

Step 4: Unable to restock: The system fails to update the inventory due to a system error and displays an “Inventory Update Failed” message. The System remains on the “Manage Inventory” page and the staff member is allowed to try again.

Use Case Name: Reply to Customer messages

Summary: Staff members view and respond to customer messages to answer questions and resolve issues. The system records each response and marks the original message as acknowledged.

Actors: Staff

Precondition: Staff is logged into the system and there are unread customer messages in the queue.

Description of main sequence:

1. Staff selects the option to view the customer message queue.
2. The system displays all unread customer messages.
3. Staff selects a message from the queue to open and view its content.
4. Staff composes and sends a reply.
5. The system records the response and sends the message.
6. The system marks the original message as read.

Postcondition: The customer's message is marked as read and a reply is sent.

Alternative sequences:

Step 1: Staff is unable to view the queue of customer messages: A system error prevents the Staff from viewing customer messages. The system displays a “Something Went Wrong. Try Again Later” message. Staff remain on the same screen and have the option to reload the page.

Step 3: No unread messages: There are no unread messages in the queue. The system displays “No New Messages” message.

Step 4: Message not delivered: The staff member attempts to respond to a message but the system fails to deliver it. The message content remains in the input field and staff can retry sending the message.

Step 4: Maximum words reached : The system limits the user to write the message at a certain words limit. If the user tries to exceed it, the system informs the user about the maximum limit reached.

Step 4: Minimum words not met : At Step 3, if the customer attempts to send a message that is below the minimum word limit , the system prevents the message from being sent and displays an alert message.

Use Case Name: Generate daily Report

Actors: System timer

Summary: The timer triggers the system to generate a report at 9pm every business day. The report contains a summary of all sales made that day. Each report includes the product's name and quantity sold and is automatically sent to the CEO for review.

Precondition: It is 9pm on a given business day and sales data for the current day is available.

Description of main sequence:

1. The timer triggers the report generation process.
2. The system generates the total sales data for the current business day.
3. The system records all products sold today, name and its quantity sold.
4. The system aggregates all this data into one daily report and stores it.
5. At the end of each month the system compiles the amount of products sold every day and week along with the name of each product.
6. The daily report and monthly reports are separated and sent to the CEO.

Postcondition: The daily report is generated, stored, and delivered to the CEO.

Alternative sequences:

Step 2: No sales: If no sales occurred that day, the system still generates the report with zero sales across all products.

Step 3: Data retrieval failure: The system encounters issues when retrieving the sales data and proceeds to generate a partial report with the available data.

Step 5: The system fails to store the report due to storage unavailability: The report is discarded and the system continues running as normal.

Use Case Name: Generate Monthly Report

Actors: System timer

Summary: At 9pm on the last day of each month, the timer triggers the generation of a monthly sales report. The monthly report includes information from all available daily report summaries. The report lists each product's name, quantity sold and is automatically sent to the CEO for review.

Precondition: It is the last day of the calendar month and the daily reports for the current month are available.

Description of main sequence:

1. The timer triggers the report generation process.
2. The system aggregates all daily sales for the current month
3. The system then compiles the data into one monthly report and stores it in the designated location.

Postcondition: The monthly generated and stored.

Alternative sequences:

Step 2: Unable to aggregate report: The system is unable to aggregate all daily reports for the month and summarize them. The system will continue only with available data.

Step 2: No sales: If no sales occurred that month, the system still generates the report with zero figures.

Step 3: The System fails to store the report due to storage unavailability. The report is not saved and the system continues running as normal.

Use Case Name: View Report

Actors: CEO

Summary: CEO access and views the daily and the monthly sales reports.

Precondition: CEO has logged in and the system has already generated the reports.

Description of main sequence:

1. The CEO has logged into the system.
2. The CEO selects the reports section.
3. The system displays the list of already generated sales reports, both monthly and daily.
4. The CEO selects the report to view it.
5. The system displays the contents of the selected sales report.

Postcondition: The CEO has viewed the sales report.

Alternative sequences:

Step 2: No reports found: System displays the message “No Reports Found ” and remains on the “View Report” page.

Step 5: The system fails to load a report: The system fails to display the selected report, displays “Something Went Wrong. Try Again Later or Select a Different Report” message, and return to the “View Report” page.

Cruz, Andrea V.	acruz189406@troy.edu
Brooks, Vincent P.	vbrooksjr@troy.edu
Patel, Meghkumar H.	mpatel224694@troy.edu